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【 CASE REPORT 】

Acute onset of constrictive pericarditis due to acute myelomonocytic leukemia: A case and literature review

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Abstract:

We herein present a fatal case of constrictive pericarditis (CP) due to acute myelomonocytic leukemia (AMML) in a patient who initially complained of an acute onset of chest pain two days after COVID-19 vaccination. An autopsy revealed pericardial infiltration of leukemic cells. CP is rarely associated with leukemia and only 14 cases have been reported in the literature. The etiology of CP in previous reports included leukemic infiltration, graft-versus-host disease, drug-induced, post-radiation, autoimmune, and otherwise unidentified. This case indicates that leukemic infiltration can cause CP and that clinicians should include leukemia in the differential diagnosis of CP.

Key words: Acute pericarditis, Constrictive pericarditis, AMML, AML with NUP98 translocation

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Introduction

Constrictive pericarditis (CP) is a rare disease, and its true prevalence in the population is unknown. According to existing reports, most cases of CP are idiopathic, and it is often difficult to identify the cause of CP. Cardiac surgery, radiation therapy, malignancy, and infection are the next most common causes; however, the prognosis for each setting is unclear. We herein present a fatal case of acute myelomonocytic leukemia (AMML) with CP.

Case Report

A 64-year-old female developed chest pain upon inhalation. She had been followed up at the Department of Respiratory Medicine for small cell lung carcinoma, which was diagnosed 26 months prior to this presentation, and had already undergone a 3rd line of chemotherapy (1st line: cisplatin, etoposide, irradiation; 2nd line: cisplatin, irinotecan; 3rd line: amrubicin). She had received a COVID-19 mRNA vaccine two days before the onset of her chest pain. The patient was worried about vaccine-induced pericarditis. Echocardiography and computed tomography (CT) showed circumferential pericardial effusion and thickening (Fig. 1a, b), and she was referred and admitted to the Department of Cardiology with a suspicion of pericarditis. Chest radiography showed a slight enlargement of the heart and right pleural effusion. Electrocardiography showed a low voltage without ischemic changes. Laboratory tests revealed an increased white blood cell count with monocytosis, anemia, and thrombocytopenia (Table 1). Her complete blood count was

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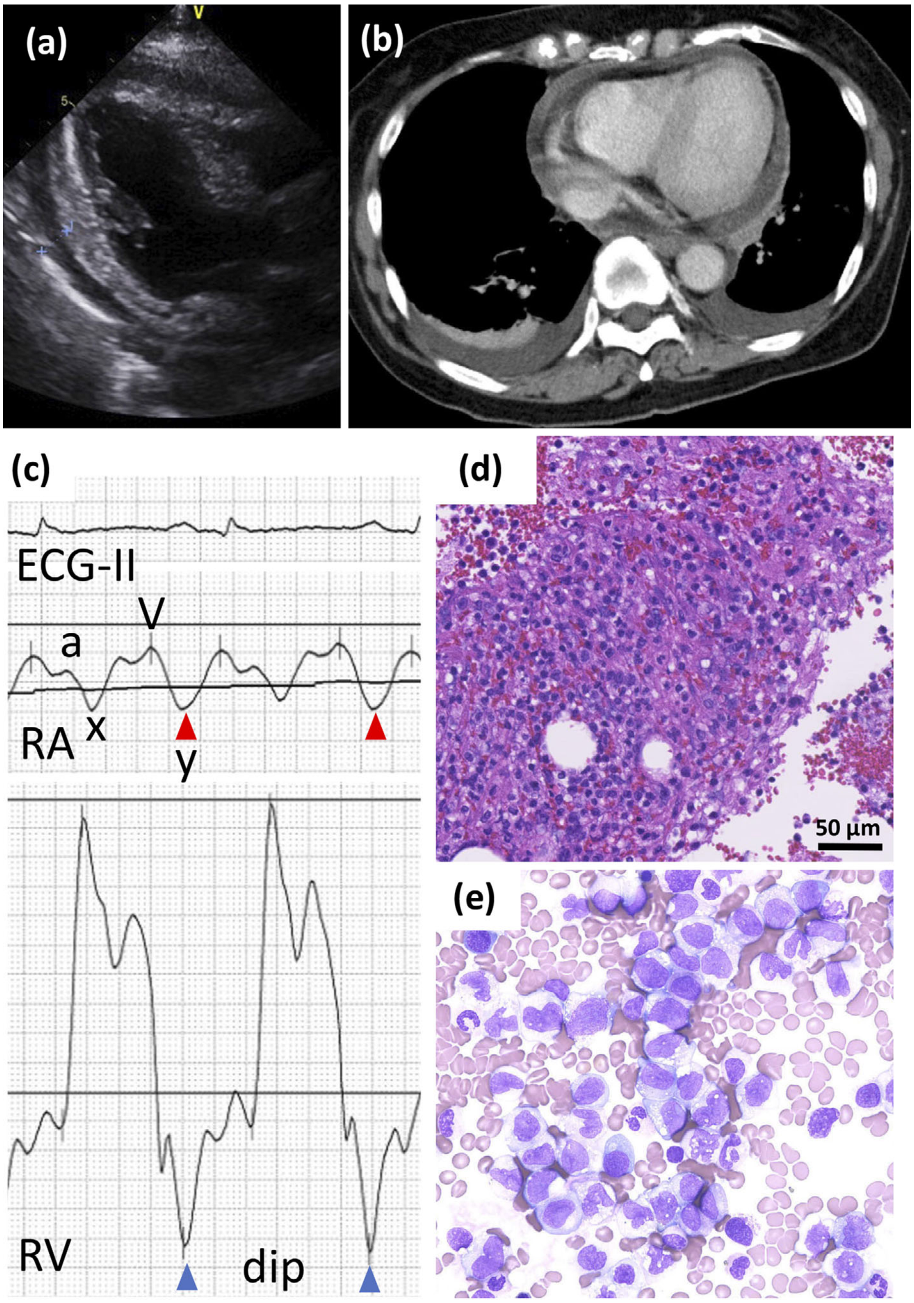

Figure 1. **Constrictive pericarditis and leukemia. (a) Transthoracic echocardiogram. Pericardial effusion was observed around the entire heart. (b) Chest CT. Pericardial thickening and enhancement were observed with pericardial effusion. Bilateral pleural effusion was also seen. (c) Telemetry monitoring and simultaneous right atrium (RA) and right ventricular (RV) pressure recordings after pericardiocentesis. Sharpened y descent of the right atrium (red arrowhead) and dip of the right ventricle (blue arrowhead) were observed. This is a consistent finding of constrictive pericarditis. (d) Cell block sections of the pericardial effusion showed mononuclear cell infiltration. Hematoxylin and Eosin staining. (e) Bone marrow smear. Diffuse proliferation of medium-sized atypical cells with frosted nuclei and well-defined nucleoli, with a few normal hematopoietic cells present. Erythroblasts and megakaryocytic cells were rarely observed.**

unremarkable when she visited the Department of Respiratory Medicine 17 days prior to admission. Aspirin and colchicine were administered for pericarditis, but the pleural effusion increased and the urine volume decreased. Therefore, pericardiocentesis was performed on day 9 as a treatment and diagnostic procedure. After the drainage of 150 mL of pericardial effusion, right heart catheterization showed a sharp descent in the right atrium and a dip pattern in the right ventricle, representing CP (Fig. 1c). Cytology of a pericardial effusion specimen showed increased histiocytes and lymphocytes, and was reported as reactive inflammation (Fig. 1d). No evidence of lung carcinoma infiltration was

Table 1. Laboratory Data on Admission.

WBC	88,600 /μL	TP	5.7 g/dL	IgG	640 mg/dL
neutro	44.5 %	ALB	3.3 g/dL	IgA	70 mg/dL
lymph	4.0 %	T-Bil	0.8 mg/dL	IgM	35 mg/dL
mono	44.5 %	AST	23 U/L	ANA	<40 times
eosino	0.0 %	ALT	8 U/L	RF	<4.5 IU/mL
baso	0.0 %	LD	775 U/L	NT-proBNP	1,576 pg/mL
metamyelo	2.5 %	ALP	91 U/L		
myelo	4.0 %	γ-GTP	44 U/L		
promyelo	0.5 %	BUN	14 mg/dL		
blast	0.0 %	Cr	0.75 mg/dL		
RBC	3.64×10^6 /μL	Na	140 mEq/L		
Hb	11.3 g/dL	K	3.0 mEq/L		
Plt	21×10^3 /μL	Cl	103 mEq/L		
		Ca	8.6 mg/dL		
PT	13.2 sec	TG	84 mg/dL		
APTT	37.6 sec	LDL	39 mg/dL		
Fbg	598 mg/dL	CRP	12.5 mg/dL		
FDP	10.4 μg/dL				
D-dimer	3.03 mg/dL				

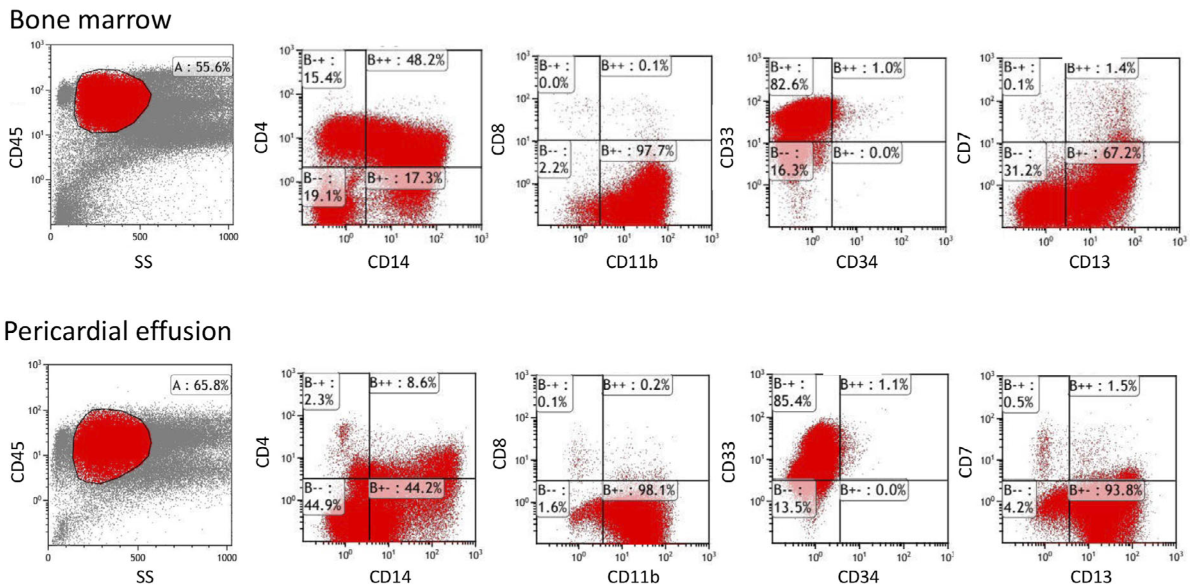

Figure 2. Flow cytometry of bone marrow and pericardial effusion.

found. The effusion culture result was negative.

She was referred to the hematology department on the day of admission with an increased WBC count. Bone marrow aspiration revealed hypercellular marrow with an increased number of immature monocytes and dysplastic granulocytes (Fig. 1e). Typical myeloblasts did not increase in number and they appeared as myeloproliferative neoplasms. JAK2, CALR, and FLT3 mutations were negative. G-banding showed 46,XX and added (11)(p15)[19/20]. A NUP98 split signal was observed in 97% of peripheral blood by fluorescence in situ hybridization analysis. Flow cytometry (FCM) of the bone marrow was positive for CD4, 11b, 13, 14, and 33, but negative for CD34. FCM of pericardial effusion showed the same pattern as bone marrow except for CD4 expression, which was lower in pericardial effusion (Fig. 2). A diagnosis of acute myelomonocytic leukemia (AMML) in FAB classification/AML post-cytotoxic therapy according to the WHO 5th classification (1) / AML with other rare recurring translocations, therapy-related in the International Consensus Classification 2022 classification (2), was made. The pericardial effusion in this case was due to leukemic pericarditis. The patient was then transferred to the hematology department. The clinical course of the patient is shown in Fig. 3. Due to the patient's poor general status, treatment with hydroxyurea followed by azacitidine was started to debulk the leukemic cells, and we planned to thereafter perform venetoclax treatment. However, she died of a sudden brain hemorrhage on day 4 of

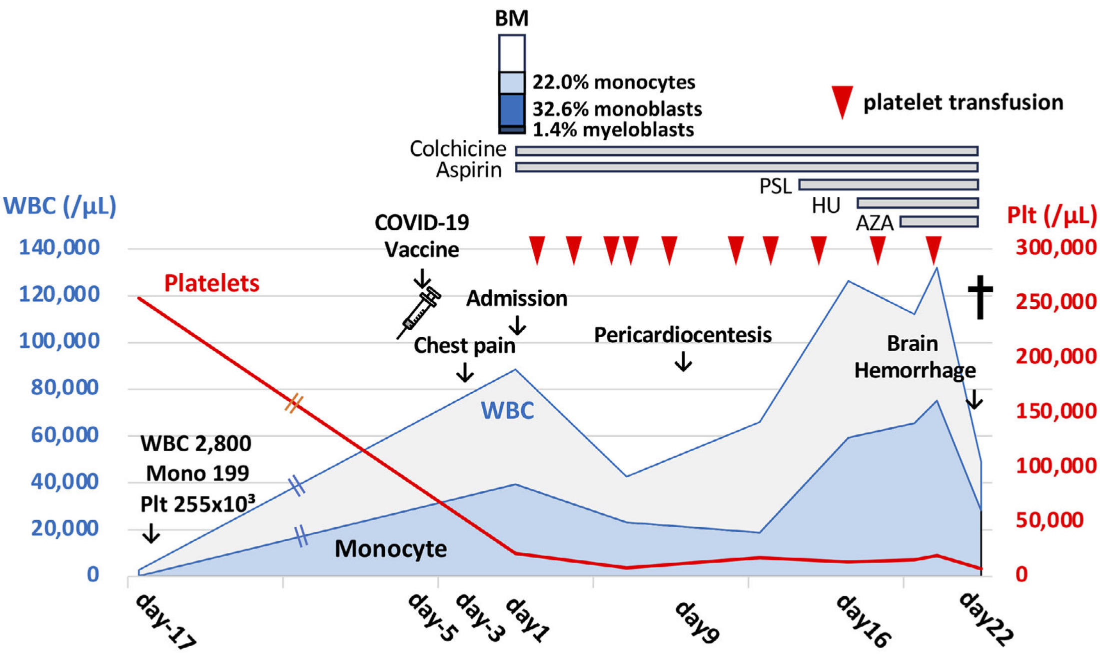

Figure 3. **Clinical course. AZA: azacitidine, HU: hydroxyurea, BM: bone marrow, Mono: monocyte, Plt: platelet, PSL: prednisolone, WBC: white blood cell**

azacitidine treatment. Laboratory data revealed evidence of disseminated intravascular coagulation (FDP 44.7 μg/mL, D-dimer 30.04 μg/mL, fibrinogen 99 mg/dL; PT-INR, 1.53, AT-3 68%). CT revealed an extensive cerebral hemorrhage and herniation at the foramen magnum. An autopsy was performed with the consent of her family.

Autopsy findings

The vertebra showed red marrow (Fig. 4a) and it was filled with leukemic cells, showing almost 100% cellularity with a few normal hematopoietic cells (Fig. 4b). Fibrinous pericarditis was observed on the heart surface (Fig. 4c). Microscopically, leukemic cells infiltrated the epicardium of the left ventricle with massive fibrin exudates (Fig. 4d, e). The leukemic cells were immunohistochemically positive for both MPO and CD68 (PGM-1), suggesting the infiltration of AMML (Fig. 4f, g). AE1/AE3 showed no evidence of lung cancer metastasis. Hemorrhaging was observed in multiple organs, especially in the pancreas and esophagus. The primary site of small cell lung carcinoma, the hilum of the left lung, showed extensive fibrosis and no residual cancer cells. A craniotomy was not performed.

Discussion

In Western Europe, most cases of underlying disease-causing CP are idiopathic (78-86%), with the remainder being neoplastic (5-7%), tuberculous (4%), and autoimmune or post-cardiac surgery (1-7%) (3). The risk of progression to CP is low (<1%) in viral and idiopathic pericarditis, intermediate (2-5%) in immune-mediated pericarditis and neoplastic pericardial diseases, and high (20-30%) in bacterial pericarditis (4). The standard treatment for idiopathic pericarditis is aspirin and colchicine (3); however, our patient did not respond to this treatment.

Clinical cases of CP with leukemia, as in our case, are very rare, and their characteristics have not been clarified. Therefore, we searched for case reports of CP associated with leukemia on PubMed. We found 13 case reports and 14 clinical cases when we searched with the terms “(leukemia) AND (constrictive pericarditis)” (Table 2) (5-17). Male patients were remarkably predominant in comparison to female patients (13 male cases and 1 female case). Nine (64.3%) of the 14 patients had myeloid leukemia. Of these nine patients, four (44%) had monocytic leukemia. Common symptoms of suspected CP were dyspnea (10/14 cases), followed by fatigue and edema. Pericardial effusion was observed in 8 cases. Six cases resulted in fatal outcomes mainly due to cardiac arrest. Surgical intervention with pericardiectomy was performed in 8 patients. Colchicine and steroids are commonly used non-surgical interventions that can potentially ameliorate the symptoms of CP. The causes of CP were leukemic infiltration (N=4), graft-versus-host diesease (N=3), drug-induced (N=2), autoimmune (N=1), radiation (N=1), and unidentified (N=3) (Table 2).

We initially thought that the increased WBC count in the peripheral blood was a reactive increase due to pericarditis because our patient had a normal blood count just 3 weeks before admission. However, a rapid increase in immature monocytes and NUP98 translocation clearly indicate hematological malignancy. Initial cytology of the pericardial effusion suggested inflammatory cell infiltration due to rack of CD34 positive population, however the FCM pattern of the pericardial effusion matched that of the leukemic bone mar-

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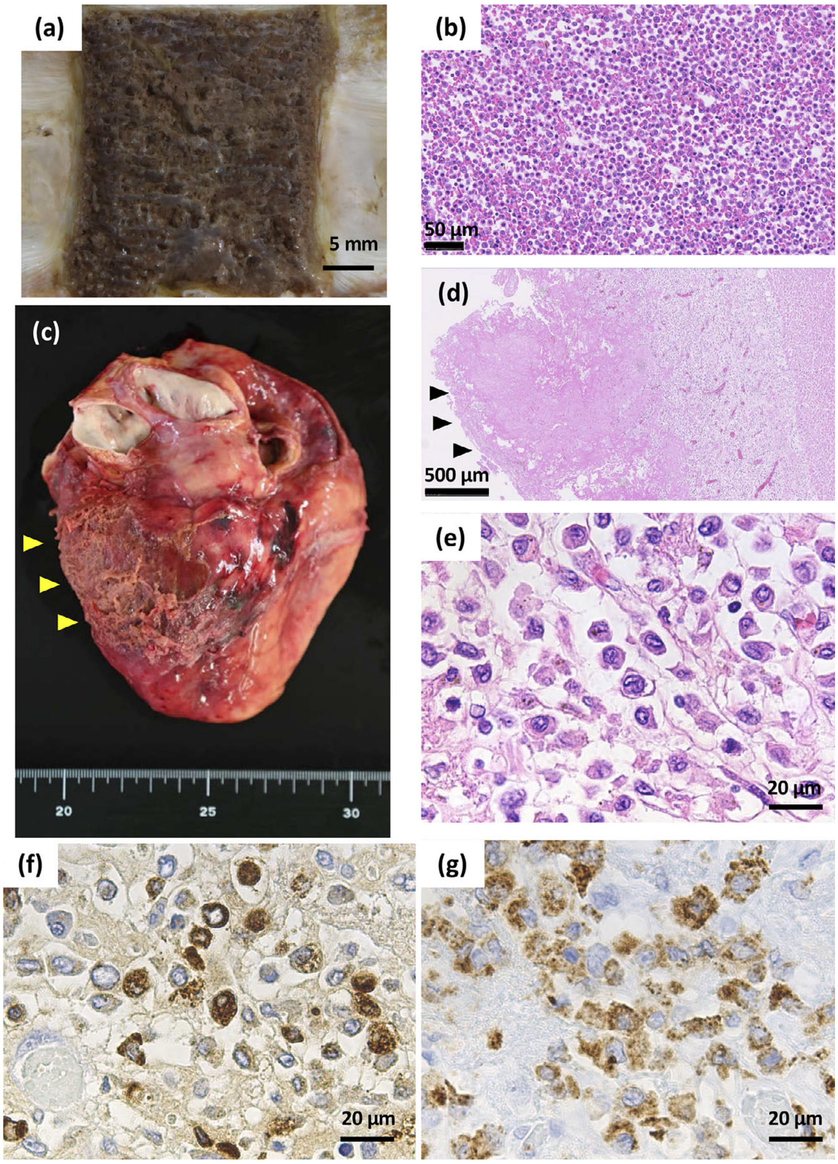

Figure 4. Pathological findings. (a) A macroscopic view of the vertebra showed red marrow. (b) High magnification of bone marrow of the vertebra showed hypercellular marrow with diffuse proliferation of leukemic cells. Hematoxylin and Eosin (H&E) staining. (c) A macroscopic view of the heart showed fibrinous pericarditis on the anterior to posterior surface of the left ventricle (yellow arrowheads). (d) Low magnification of the epicardium showed fibrin exudates (black arrowheads) and leukemic cell infiltration. H&E staining. (e) High magnification of the epicardium showed infiltration of leukemic cells with irregular and convoluted nuclei. H&E staining. (f) Myeloperoxidase (MPO) and (g) CD68 stain of the epicardium. Infiltrating leukemic cells were positive for MPO and CD68.

row. Finally, the patient was diagnosed with leukemic constrictive pericarditis.

Our patient was worried about the possibility of vaccine-induced pericarditis because there was increased recognition and warning about pericarditis caused by COVID-19 vaccines at that time. However, most of the reported cases of vaccine-induced pericarditis occur in young individuals (18, 19). According to data from the Japanese joint council in the Ministry of Health, Labour and Welfare, pericarditis caused by vaccination was predominant in young males (11.6-129.6 cases / million) and extremely rare in Japanese females in their 60s (0.3-5.7 cases / million) (18, 19). Furthermore, vaccine-induced pericarditis should be diagnosed after excluding other identifiable causes of pericarditis (20). One cause of CP is radiation therapy, accounting for 9-31% of all CP cases in developed countries (3). Although our patient had a history of irradiation for lung cancer, the irradiation field involved only a small part of her heart.

We herein described a rare case of CP secondary to monocytic leukemia. CP is a rare complication of the initial presentation of hematological malignancies. Leukemic infil-

Table 2. Review of the Literature on Constrictive Pericarditis in Cases of Leukemia.

Case No.	Age	Sex	Hematological diagnosis	Treatment for leukemia	Symptoms	Pericardial effusion	Cause of pericarditis	Treatment for pericarditis	Outcome	Cause of death
1[5]	55	F	CLL	-	dyspnea, anginal pain, edema	-	leukemic infiltration	pericardiectomy	dead (autopsy)	cardiac arrest
2[6]	47	M	AML (M2)	DNR+AraC	fatigue, hemorrhage, dyspnea	+	chemotherapy	pericardiocentesis	alive	
3[7]	43	M	ALL (Ph+)	HSCT with TBI12Gy	dyspnea	+	GVHD	pericardiectomy	alive	
4[7]	39	M	ALL (Ph+)	HSCT with TBI12Gy	dyspnea	-	radiation	pericardiectomy	dead	septic shock
5[8]	43	M	AML (M5)	-	dyspnea, orthopnea, abdominal distension	+	leukemic infiltration	-	dead (autopsy)	cardiorespiratory arrest
6[9]	42	M	PEL/CML	HSCT with TBI	chest pain, lethargy, nausea, weight loss, edema	-	GVHD and/or viral pericarditis	pericardiectomy, high-dose steroid	dead	cardiac failure
7[10]	64	M	CML	nilotinib	unknown	-	nilotinib	pericardiectomy	unknown	
8[11]	57	M	CLL/SLL	bendamustine, rituximab	fever, night sweat	+	leukemic infiltration	prednisone, azathioprine, pericardiectomy	alive	
9[12]	late 70s	M	CMML	-	epigastric pain, weight loss	-	autoimmune	pericardiectomy, colchicine, steroids, IVIG	alive	
10[13]	54	M	AML (M4) recurrence	HSCT	dyspnea, edema	+	leukemic infiltration	chemotherapy	dead	leukemia
11[14]	60	M	MDS/AML	HSCT	dyspnea following viral-like syndrome	+	not identified	colchicine	alive	
12[15]	74	M	AML	HSCT	dyspnea, edema	+	GVHD	pericardetomy, ruxolitinib	alive	
13[16]	55	M	Erdheim-Chester disease/CMML	-	fatigue, dyspnea	-	not identified	corticosteroid	alive	
14[17]	39	M	AML	DNR+AraC	epigastric discomfort, dyspnea, headache	+	not identified	pericardiocentesis	dead	cardiac arrest, tamponade
Our case	64	F	AMML (FAB) / AML-pCT (WHO 5th)/ AML with other rare recurring translocations (ICC2022)	HU, AZA	chest pain, dyspnea	+	leukemic infiltration	colchicine, PSL	dead (autopsy)	DIC, brain hemorrhage

M: male, F: female, ALL: acute lymphoblastic leukemia, AML: acute myeloid leukemia, AMML: acute myelomonocytic leukemia, AraC: cytarabine, AZA: azacitidine, CLL: chronic lymphocytic leukemia, CML: chronic myeloid leukemia, CMML: chronic myelomonocytic leukemia, DIC: disseminated intravascular coagulation, DNR: daunorubicin, GVHD: graft-versus-host disease, HSCT: hematopoietic stem cell transplantation, HU: hydroxyurea, ICC: International Consensus Classification, IVIG: intravascular immunoglobulin, MDS: myelodysplastic syndrome, pCT: post cytotoxic chemotherapy, PEL: primary effusion lymphoma, Ph: Philadelphia chromosome, PSL: prednisolone, SLL: small lymphocytic lymphoma, TBI: total body irradiation, WHO: World Health Organization

tration should be considered as a cause of CP, especially if the patient shows abnormal blood counts.

The authors state that they have no Conflict of Interest (COI).

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